

Webcam Home Security System



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Here is a simple webcam interface program for home security application. This security solution can be used as a motion detector in shops, office buildings and other commercial setups. It is very inexpensive compared to CCTV cameras and the webcam with USB cable can be easily installed in any laptop or PC. Also, up to four webcams can be used, and the captured file stored in bmp format in a selected folder in your system.

Software program

The software is written in Visual C#. It allows user to do high-end tasks like enabling camera and do the image processing over Visual Basic. It is a very high-speed and sophisticated environment that enables the user to design software for:

1. Image processing
2. Video processing
3. Analysis of different images
4. Contour analysis in frame
5. Pattern analysis
6. Image comparison

The project uses C# along with AForge application tool.

AForge.NET is a C# Framework designed for developers and researchers in the fields of computer vision and artificial intelligence for image processing, neural networks, genetic algorithms, machine learning, robotics, etc. The whole setup requires installation of dotnet Framework 4.0 or higher version.

The Framework comprises the following set of libraries and

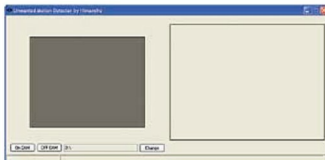


Fig. 1: Program output without webcam



Fig. 2: An image being captured by webcam



Fig. 3: The motion being captured by webcam

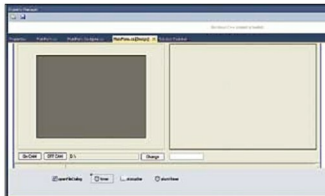


Fig. 4: MainFrame screenshot during development stage

sample applications:

1. AForge.Imaging: Library with image processing routines and filters
2. AForge.Vision: Computer vision library
3. AForge.Video: A set of libraries for video processing
4. AForge.Neuro: Neural networks computation library
5. AForge.Genetic: Evolution programming library
6. AForge.Fuzzy: Fuzzy computations library
7. AForge.Robotics: A library providing support of some robotics kits
8. AForge.Machine-Learning: Machine learning library

MainFrame.cs is the main program in this application. Its screenshot is shown in Fig. 4. The program captures the image every one second whenever motion is detected by the webcam.

You can change the image capture time while designing MainFrame.cs.

For this, first select 'timer' option at the bottom left in MainForm.cs. Right-click it and choose property of the timer. By default, the interval value is set to 500. You can change the interval value from 500 to 200. This will change the image capture time from 500 ms to 200 ms. So more images will be captured. If you increase the interval value to 1000, fewer images will be